ICCS206 Discrete Mathematic: Homework 5

Due on Sunday, October 24, 2021

Pakawut Jiradilok Section 1

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Problem 1

Find closed-form expressions for the following sums/products.

1. $\sum_{i=1}^{n}(2i+1)$
2. $\sum_{i=1}^{n}\sum_{j=1}^{m}(i+j^2)$
3. $\sum_{i=1}^{n}\sum_{j=1}^{m}2^{i+2j}$
4. $\prod_{i=1}^{n}\prod_{j=1}^{m}2^i3^j$

Solution

1.

$$\sum_{i=1}^{n}(2i+1) = \sum_{i=1}^{n}2i + \sum_{i=1}^{n}1 = 2\sum_{i=1}^{n}i + \sum_{i=1}^{n}1 = 2\left(\frac{n(n+1)}{2}\right) + n$$

$$\boxed{2\left(\frac{n(n+1)}{2}\right) + n}$$

2.

$$\begin{aligned}
\sum_{i=1}^{n}\sum_{j=1}^{m}(i+j^2) &= \sum_{i=1}^{n}\left((i+1^2)+(i+2^2)+\cdots+(i+m^2)\right) \\
&= \sum_{i=1}^{n}\left(im+(1^2+2^2+3^2+\cdots+m^2)\right) \\
&= 1m(1^2+2^2+3^2+\cdots+m^2)+2m(1^2+2^2+3^2+\cdots+m^2)+...+nm(1^2+2^2+3^2+\cdots+m^2) \\
&= (1^2+2^2+\cdots+m^2)\,(1m+2m+\cdots+nm) \\
&= \left(\frac{m(m+1)(2m+1)}{6}\right)\cdot m\cdot\left(\frac{n(n+1)}{2}\right)
\end{aligned}$$

$$\boxed{\left(\frac{m(m+1)(2m+1)}{6}\right)\cdot m\cdot\left(\frac{n(n+1)}{2}\right)}$$

3.

$$\begin{aligned}\sum_{i=1}^{n}\sum_{j=1}^{m} 2^{i+2j} &= \sum_{i=1}^{n}\left(2^{i+2(1)} + 2^{i+2(2)} + 2^{i+2(3)} + \cdots + 2^{i+2m}\right)\\ &= \sum_{i=1}^{n}\left(2^{i}\cdot 2^{2} + 2^{i}\cdot 2^{4} + 2^{i}\cdot 2^{6} + \cdots + 2^{i}\cdot 2^{2m}\right)\\ &= \sum_{i=1}^{n} 2^{i}\left(2^{2(1)} + 2^{2(2)} + 2^{2(3)} + \cdots + 2^{2m}\right)\end{aligned}$$

let

$$S = 2^{2(1)} + 2^{2(2)} + 2^{2(3)} + \cdots + 2^{2m}$$

, and

$$2^2 S = 2^2 + 2 + 2^{2(2)} + 2 + 2^{2(3)} + 2 + \cdots + 2^{2m} + 2$$

subtract the two

$$2^2 S - S = 2^{2m+2} - 2^{2(1)} \Rightarrow S = \frac{2^{2m+2} - 2^2}{2^2 - 1}$$

then, we continue

$$\begin{aligned}\sum_{i=1}^{n} 2^{i}\left(\frac{2^{2m+2}-4}{3}\right) &= 2^{1}\left(\frac{2^{2m+2}-4}{3}\right) + 2^{2}\left(\frac{2^{2m+2}-4}{3}\right) + \cdots + 2^{n}\left(\frac{2^{2m+2}-4}{3}\right)\\ &= \left(\frac{2^{2m+2}-4}{3}\right)\left(2^{1} + 2^{2} + \cdots + 2^{n}\right)\end{aligned}$$

let $S' = 2^1 + 2^2 + 2^3 + \cdots + 2^n$, and $2S' = 2^2 + 2^3 + 2^4 + \cdots + 2^{n+1}$, we find

$$2S' - S' = 2^{n+1} - 2^1 \Rightarrow S' = 2^{n+1} - 2$$

and therefore

$$\boxed{\left(\frac{2^{2m+2}-4}{3}\right)\left(2^{n+1} - 2\right)}$$

4.

$$\begin{aligned}\prod_{i=1}^{n}\prod_{j=1}^{m} 2^{i}\cdot 3^{j} &= \prod_{i=1}^{n}\left((2^{i}\cdot 3^{1})(2^{i}\cdot 3^{2})(2^{i}\cdot 3^{3})\cdots(2^{i}\cdot 3^{m})\right)\\ &= \prod_{i=1}^{n}\left(2^{im}\cdot 3^{\frac{m(m+1)}{2}}\right)\end{aligned}$$

$$= \left(2^m \cdot 3^{\frac{m(m+1)}{2}} \cdot 2^{2m} \cdot 3^{\frac{m(m+1)}{2}} \cdot ... \cdot 2^{nm} \cdot 3^{\frac{m(m+1)}{2}}\right)$$
$$= \left(3^{\frac{m(m+1)}{2}} \cdot 2^{(m+2m+3m+...+nm)}\right)$$

$$\boxed{\left(3^{\frac{m(m+1)}{2}}\right)^n \cdot 2^{m\left(\frac{n(n+1)}{2}\right)}}$$

Problem 2

Use integral bounds to give upper and lower bounds for

1. $\displaystyle\sum_{x=1}^{n} \frac{1}{x^2}$
2. $\displaystyle\sum_{x=1}^{n} x^{3/2}$

Solution

1. $\displaystyle\sum_{x=1}^{n} \frac{1}{x^2}$ we let $f(x) = \frac{1}{x^2}$, this is a decreasing function so

$$\int_1^{n+1} f(x)\,dx \leq \sum_{k=1}^{n} f(k) \leq \int_1^{n} f(x)\,dx + f(1)$$

find the integral first

$$\int f(x)\,dx = \int \frac{1}{x^2}\,dx = \int x^{-2}\,dx$$
$$= x^{-}1 + c$$

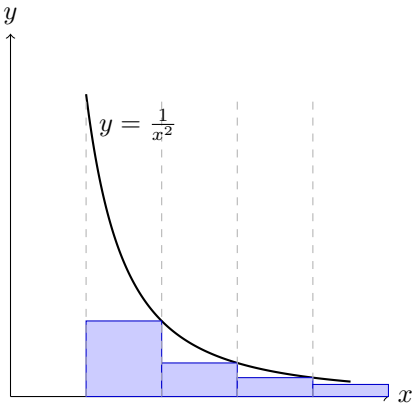
then we can calculate the bound

$$\left(-\frac{1}{x}\right)\Bigg|_1^{n+1} \leq \sum_{k=1}^{n} f(k) \leq \left(-\frac{1}{x}\right)\Bigg|_1^{n} + f(1)$$

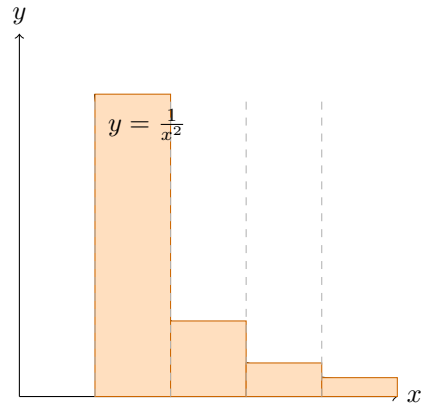
$$-\frac{1}{n+1} - (-1) \leq \sum_{k=1}^{n} f(k) \leq -\frac{1}{n} - (-1) + 1$$

$$\boxed{1 - \frac{1}{n+1} \leq \sum_{x=1}^{n} \frac{1}{x^2} \leq 2 - \frac{1}{n}}$$

The plots below shows the unit-width rectangles for both bounds.



Lower bound $\int_1^{n+1} f(x)\,dx$



Upper bound $\int_1^{n} f(x)\,dx + f(1)$

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2. $\displaystyle\sum_{x=1}^{n} x^{3/2}$ we let $f(x) = x^{3/2}$, this is an increasing function so

$$\int_1^n f(x)\,dx \le \sum_{k=1}^{n} f(k) \le \int_2^{n+1} f(x)\,dx$$

find the integral first

$$\int f(x)\,dx = \int x^{3/2}\,dx$$

$$= \frac{x^{5/2}}{5/2} + c$$

$$= \frac{2}{5}x^{5/2} + c$$

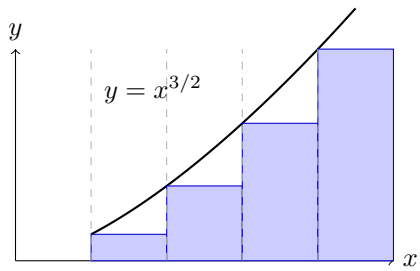
then we calculate the bound

$$\left(\frac{2x^{5/2}}{5}\right)\Bigg|_1^n \le \sum_{k=1}^{n} f(k) \le \left(\frac{2x^{5/2}}{5}\right)\Bigg|_2^{n+1}$$

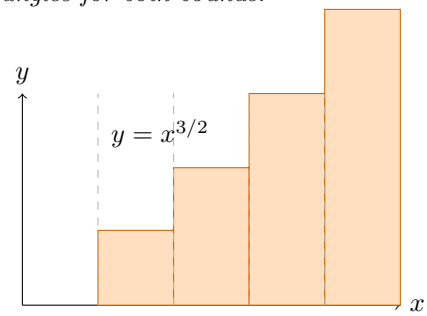
$$\frac{2n^{5/2}}{5} - \frac{2}{5} \le \sum_{k=1}^{n} f(k) \le \frac{2(n+1)^{5/2}}{5} - \frac{2\cdot 2^{5/2}}{5}$$

$$\boxed{\frac{2}{5}\left(n^{5/2} - 1\right) \le \sum_{x=1}^{n} x^{3/2} \le \frac{2}{5}(n+1)^{5/2} - \frac{2}{5}\cdot 2^{5/2}}$$

The plots below shows the unit-width rectangles for both bounds.



Lower bound $\int_1^n f(x)\,dx$



Upper bound $\int_2^{n+1} f(x)\,dx$

Problem 3

Bossy wants to build an n-story house of cards (pyramid style). Derive a formula for the number of cards required for n stories, and determine how many decks are needed for a 10-story house.

Solution

observe the given reference, we'll see that

$$\begin{aligned}
T(1) &= 2 = 2(1) + (0) \\
T(2) &= 7 = 2(1+2) + (0+1) \\
T(3) &= 2(1+2+3) + (0+1+2) = 15 \\
T(4) &= 2(1+2+3+4) + (0+1+2+3) = 26
\end{aligned}$$

which we can derive the summation as

$$T(n) = 2\sum_{i=1}^{n} i + \sum_{j=0}^{n-1} j$$

simplies to

$$T(n) = 2\frac{n(n+1)}{2} + \frac{(n-1)(n)}{2}$$

now, we find T(10) with that simplified formula as

$$T(10) = 2\frac{10(10+1)}{2} + \frac{(10-1)(10)}{2} = 155$$

we'll need $\frac{T(10)}{52} = \frac{155}{52} = \boxed{3}$ decks of card for a 10-story house.

Problem 4: Data Structure Cheat Sheet

Solve or verify the following recurrences (assume $T(1) = 1$).

1. $T(n) = T(n-1) + n$ (verify by induction).
2. $T(n) = T(n-1) + n^3$.

3. $T(n) = T(\lfloor n/2 \rfloor) + 1$.
4. $T(n) = T(\lfloor n/2 \rfloor) + n$.
5. $T(n) = 2T(\lfloor n/2 \rfloor) + n$.
6. $T(n) = 2T(\lfloor n/2 \rfloor) + 1$.
7. (Optional) $T(n) = 2T(\lfloor n/2 \rfloor) + \log_2 n$.

Solution

1. $T(n) = T(n-1) + n$; $T(1) = 1$

   we expand the recurrence to observe the accumulating pattern, then summarize the resulting series to express the closed-form solution.

$$\begin{aligned}
T(1) &= 1 \\
T(2) &= T(1) + 2 = 1 + 2 = 3 \\
T(3) &= T(2) + 3 = 1 + 2 + 3 = 6 \\
T(n) &= T(n-1) + n = 1 + 2 + 3 + \cdots + (n-1) + n \\
&= \sum_{i=1}^{n} i \\
\text{T(n)} &= \frac{n(n+1)}{2}
\end{aligned}$$

   now we prove $T(n) = \dfrac{n(n+1)}{2}$ with induction

   the claim

$$P(n) = T(n) = \frac{n(n+1)}{2}$$

   for the base case where $n_0 = 1$

$$\begin{aligned}
P(1) = T(1) &= \frac{1(1+1)}{2} = 1 \\
P(2) = T(2) &= \frac{2(3)}{2} = 3 \\
P(3) = T(3) &= \frac{3(4)}{2} = 6
\end{aligned}$$

   we assume $P(k)$ for arbitrary $k \geq n_0$, then try to prove $P(k+1) \equiv (k+1)(k+2)/2$

$$\begin{aligned}
P(k+1) &= P(k) + (k+1) \\
&= \frac{k(k+1)}{2} + \frac{2(k+1)}{2} \quad \text{by Inductive Hypothesis} \\
&= \frac{(k+1)(k+2)}{2}
\end{aligned}$$

   since $P(k+1)$ comes after $P(k)$, our induction case is complete, and we can conclude the claim is true.

$$\boxed{T(n) = \frac{n(n+1)}{2}}$$

---

2. $T(n) = T(n-1) + n^3$; $T(1) = 1$

   we expand the recurrence to observe the accumulating pattern, then summarize the resulting series to express the closed-form solution.

$$\begin{aligned}
&T(1) = 1 \\
&T(2) = T(1) + 2^3 = 1 + 2^3 \\
&T(3) = T(2) + 3^3 = (1 + 2^3) + 3^3 \\
&\quad\vdots \\
&T(n) = T(n-1) + n^3 = \sum_{i=1}^{n} i^3 \\
&T(n) = \left(\frac{n(n+1)}{2}\right)^2 \qquad\qquad \text{by the sum of cube formula}
\end{aligned}$$

$$\boxed{T(n) = \left(\frac{n(n+1)}{2}\right)^2}$$

---

3. $T(n) = T\left(\frac{n}{2}\right) + 1$; $T(1) = 1$

   we expand the recurrence to observe the accumulating pattern, then summarize the resulting series to express the closed-form solution.

$$\begin{aligned}
T(2^0) &= 1 = T(2^0) \\
T(2^1) &= T(1) + 1 = 1 + 1 = 2 = T(2^1) \\
T(2^2) &= T(2^1) + 1 = 2 + 1 = 3 \\
T(2^3) &= T(2^2) + 1 = 3 + 1 = 4 \\
&\vdots \\
T(2^{n'}) &= n + 1
\end{aligned}$$

   let $n = 2^{n'} \rightarrow n' = \log_2 n$ which gives

$$\boxed{T(n) = \log_2 n + 1}$$

---

4. $T(n) = T\left(\frac{n}{2}\right) + n$; $T(1) = 1$

   we expand the recurrence to observe the accumulating pattern, then summarize the resulting series to express the closed-form solution.

$$\begin{aligned}
T(2^0) &= T(1) = 1 \\
T(2^1) &= T(2^0) + 2^1 = 1 + 2 = 3 \\
T(2^2) &= T(2^1) + 2^2 = 3 + 4 = 7 \\
&\vdots \\
T(2^n) &= T(2^{n-1}) + 2^n = \sum_{i=0}^{n} 2^i
\end{aligned}$$

   we let

$$S = 2^0 + 2^1 + 2^2 + \cdots + 2^n \tag{i}$$

   then

$$\begin{aligned}
2S &= 2^1 + 2^2 + 2^3 + \cdots + 2^{n+1} \qquad \text{(ii)} \\
(ii) - (i) &: S = 2^{n+1} - 1
\end{aligned}$$

   therefore $T(2^n) = 2^{n+1} - 1$, let $n' = 2^n \to n = \log_2 n'$ which gives

$$\boxed{T(n) = 2^{\log_2 n + 1} - 1}$$

---

5. $T(n) = 2T\left(\frac{n}{2}\right) + n$; $T(1) = 1$

   we expand the recurrence to observe the accumulating pattern, then summarize the resulting series to express the closed-form solution.

$$\begin{aligned}
T(2^0) &= 2^0 = 1 \\
T(2^1) &= 2T(2^0) + 2^1 = 2 \cdot 2^0 + 2^1 = 2^1 + 2^1 = 2(2^1) \\
T(2^2) &= 2T(2^1) + 2^2 = 2(2^1 + 2^1) + 2^2 = 2^2 + 2^2 + 2^2 = 3(2^2) \\
T(2^3) &= 2T(2^2) + 2^3 = 2(2^2 + 2^2 + 2^2) + 2^3 = 2^3 + 2^3 + 2^3 + 2^3 = 4(2^3) \\
&\vdots \\
T(2^n) &= (n+1)(2^n)
\end{aligned}$$

   let $n' = 2^n \to n = \log_2 n'$ which gives

$$\boxed{T(n) = (\log_2 n + 1)\left(2^{\log_2 n}\right)}$$

---

6. $T(n) = 2T\left(\frac{n}{2}\right) + 1$; $T(1) = 1$

   we expand the recurrence to observe the accumulating pattern, then summarize the resulting series to express the closed-form solution.

$$
\begin{aligned}
T(2^0) &= 2^0 \cdot T(2^0) \\
T(2^1) &= 2T(2^0) + 1 = 2 \cdot 2^0 + 1 \\
T(2^2) &= 2T(2^1) + 1 = 2(2^0 + 1) + 1 = 2^2 + 2 + 1 \\
T(2^3) &= 2T(2^2) + 1 = 2(2^2 + 2 + 1) + 1 = 2^3 + 2^2 + 2 + 1 \\
&\vdots \\
T(2^n) &= \sum_{i=0}^{n} 2^i = 2^{n+1} - 1 \quad \text{(already prove in 4.4)}
\end{aligned}
$$

   let $n' = 2^n \to n = \log_2 n'$ which gives

$$\boxed{T(n) = 2^{\log_2 n+1} - 1}$$

Problem 5

For each pair (f, g), determine which of $\{\sim, o, O, \omega, \Omega, \Theta\}$ apply.

1. $f(n) = 30n + 900 \log n$, $g(n) = n$
2. $f(n) = \log n$, $g(n) = n$
3. $f(n) = (\log n)^2$, $g(n) = n$
4. $f(n) = \sqrt{n}$, $g(n) = (\log n)^{999}$
5. $f(n) = n2^n$, $g(n) = n$
6. $f(n) = n^2$, $g(n) = 1.0000000001^n$
7. $f(n) = 200{,}000$, $g(n) = 1$
8. $f(n) = 2^n$, $g(n) = 10^n$

Solution

1. $f(n) = 30n + 900 \log n$, $g(n) = n$

   we evaluate the limit of $\dfrac{f(n)}{g(n)}$ to determine all applicable relationships.

$$\lim_{x\to\infty} \frac{f(n)}{g(n)} = \lim_{x\to\infty} \frac{30n + 900 \log n}{n} = \lim_{x\to\infty} \frac{30 + 900\left(\frac{1}{n \ln 2}\right)}{1} = 30$$

$$\boxed{f \in O(g),\ f \in \Omega(g)}$$

---

2. $f(n) = \log n$, $g(n) = n$

   we evaluate the limit of $\dfrac{f(n)}{g(n)}$ to determine all applicable relationships.

$$\lim_{x\to\infty} \frac{\log n}{n} = 0$$

$$\boxed{f \in o(g),\ f \in O(g)}$$

---

3. $f(n) = \log_2 n$, $g(n) = n$

   we evaluate the limit of $\dfrac{f(n)}{g(n)}$ to determine all applicable relationships.

$$\lim_{x\to\infty} \frac{\log_2 n}{n} = \lim_{x\to\infty} \frac{1}{n \ln 2} = 0$$

$$\boxed{f \in O(g),\ f \in o(g)}$$

---

4. $f(n) = n$, $g(n) = (\log n)^{999}$

   we evaluate the limit of $\dfrac{f(n)}{g(n)}$ to determine all applicable relationships.

$$\begin{aligned}
\lim_{x\to\infty} \frac{(n)^{1/2}}{(\log n)^{999}} &= \lim_{x\to\infty} \frac{\frac{1}{2}n^{-1/2}}{999 \cdot (\log n)^{998} \cdot \frac{1}{n \ln 2}} \\
&= \lim_{x\to\infty} \frac{n \ln 2}{\frac{1}{2}\sqrt{n} \cdot 999 \cdot (\log n)^{998}} \\
&= \lim_{x\to\infty} \frac{\ln 2}{\{\text{some term with n}\}} = 0
\end{aligned}$$

$$\boxed{f \in O(g),\ f \in o(g)}$$

---

5. $f(n) = n2^n$, $g(n) = n$

   we evaluate the limit of $\dfrac{f(n)}{g(n)}$ to determine all applicable relationships.

$$\lim_{n\to\infty} \frac{n2^n}{n} = 2^n = \infty$$

$$\boxed{f \in \omega(g),\ f \in \Omega(g)}$$

---

6. $f(n) = n^2$, $g(n) = 1.000000000?^n$

   we evaluate the limit of $\dfrac{f(n)}{g(n)}$ to determine all applicable relationships.

$$\lim_{n\to\infty} \frac{n^2}{1^n} = \infty$$

$$\boxed{f \in \omega(g),\ f \in \Omega(g)}$$

---

7. $f(n) = 200{,}000$, $g(n) = 1$

   we evaluate the limit of $\dfrac{f(n)}{g(n)}$ to determine all applicable relationships.

$$\lim_{n\to\infty} \frac{200{,}000}{1} = 200{,}000$$

$$\boxed{f \in O(g),\ f \in \Omega(g),\ f \in \Theta(g)}$$

---

8. $f(n) = 2^n$, $g(n) = 10^n$

   we evaluate the limit of $\dfrac{f(n)}{g(n)}$ to determine all applicable relationships.

$$\lim_{n\to\infty} \frac{2^n}{10^n} = \lim_{n\to\infty} \frac{2^n}{2^n \cdot 5^n} = \lim_{n\to\infty} \frac{1}{5^n} = 0$$

$$\boxed{f \in o(g),\ f \in O(g)}$$

Problem 6

Solve the recurrences:

1. $T_{i+1} = 5T_i - 6T_{i-1}$ with $T_0 = 8$, $T_1 = 17$
2. $T_{i+1} = -T_i + 12T_{i-1}$ with $T_0 = 0$, $T_1 = -7$

Solution

We solve each recurrence by writing $T_i = x^i$, finding the characteristic equation, and then applying the initial conditions to obtain the constants.

1. we'd want to write the ith term of T_i in a form of x^i, so we let

$$T_i = x^i$$

thus

$$T_{i+1} = 5T_i - 6T_{i-1} \equiv x^{i+1} = 5x^i - 6x^{i-1}$$

divide the newly created equation with x^{i-1}*gives*

$$x^2 - 5x + 6 = 0$$

and we solve this characteristic equation with quadratic formula

$$\begin{aligned} x &= \frac{-(-5) \pm \sqrt{(-5)^2 - 4(1)(6))}}{2(1)} \\ x &= \frac{5+1}{2}, \frac{5-1}{2} \\ x &= 2, 3 \end{aligned}$$

the solution could be formed from 2^i and 3^i by adjusting the two constant attached to them

$$T_i = c_1 2^i + c_2 3^i$$

we find those two constant from the given T_0 and T_1 as

$$\begin{aligned} T_0 = 8 &= c_1 2^0 + c_2 3^0 \\ = 8 &= c_1 + c_2 \end{aligned} \tag{i}$$

and

$$\begin{aligned} T_1 = 17 &= c_1 2^1 + c_2 3^1 \\ = 17 &= 2c_1 + 3c_2 \end{aligned} \tag{ii}$$

solving this linear system gives $c_1 = 7$, $c_2 = 1$, and our closed form solution is complete

$$\boxed{T_{i+1} = 7 \cdot 2^{i+1} + 3^{i+1}}$$

$$\cong \boxed{T_i = 7 \cdot 2^i + 3^i}$$

---

2. we'd want to write the ith term of T_i in a form of x^i, so we let

$$T_i = x^i$$

thus

$$T_{i+1} = -T_i + 12T_{i-1} \equiv x^{i+1} = -x^i + 12x^{i-1}$$

divide the newly created equation with x^{i-1}*gives*

$$x^2 + x - 12 = 0$$

and we solve this characteristic equation with quadratic formula

$$\begin{aligned} x &= \frac{-1 \pm \sqrt{(1)^2 - 4(1)(-12)}}{2(1)} \\ x &= \frac{-1+7}{2}, \frac{-1-7}{2} \\ x &= 3, -4 \end{aligned}$$

the solution could be formed from 3^i and $(-4)^i$ by adjusting the two constant attached to them

$$T_i = c_1 3^i + c_2(-4)^i$$

we find those two constant from the given T_0 and T_1 as

$$\begin{aligned} T_0 = 0 &= c_1 3^0 + c_2(-4)^0 \\ = 0 &= c_1 + c_2 \end{aligned} \tag{i}$$

and

$$\begin{aligned} T_1 = -7 &= c_1 3^1 + c_2(-4)^1 \\ = -7 &= 3c_1 - 4c_2 \end{aligned} \tag{ii}$$

solving this linear system gives $c_1 = -1$, $c_2 = 1$, and our closed form solution is complete

$$\boxed{T_{i+1} = -1 \cdot 3^{i+1} + (-4)^{i+1}}$$

$$\cong \boxed{T_i = -1 \cdot 3^i + (-4)^i}$$

Problem 7: Roulette Trick Redux

Consider a deck with r red cards and b black cards. You always bet on red. If you win a turn you receive twice the bet. The bet-update rule is:

- After a win, bet w times the previous bet.
- After a loss, bet l times the previous bet.

1. Show that if $w + l = 2$, the result is independent of the card order.
2. Determine the total gain/loss when $w + l = 2$ (in terms of r, b, w, l and starting bet m).
3. (Optional) With 20 black and 18 red cards, choose w to maximize gain.

Solution

1. *Proof.* let m be the starting bet; after each round, the bet value (B_r) would be

$$B_{u+v} = m \cdot w^u \cdot l^v \text{ where } u, v \text{ be amount of win and lost accordingly}$$

to prove that permutation of the card doesn't matter, assume the final result of the game

$$result = B_1 + B_2 + B_3 + \cdots + B_n \text{ where } n = u + v$$

if we swap adjacent turns' result, say B_i and B_{i+1}, where $(1 \leq i \leq n)$, there'll be 4 cases as

case 1 and case 2: win-win and lost-lost — this would resulted in no changes, as win or lost isn't unique.

case 3: win-lost $\mapsto$ lost-win

let B_i be $m \cdot w^u \cdot l^v$, then B_{i+1} would be $-m \cdot w^{u+1} \cdot l^v$ in case of B_i is a win; the sum of these two adjacent terms are

$$B_i + B_{i+1} = m \cdot w^u \cdot l^v(1 - w)$$

after the swap, $B'i = -m \cdot w^u \cdot l^v$, and $B'i + 1$ would be $m \cdot w^u \cdot l^{v+1}$. the sum of these two adjacent terms are

$$B'_i + B'_{i+1} = m \cdot w^u \cdot l^v(l - 1)$$

we're given

$$w + l = 2 \mapsto l = 2 - w$$

substitude l with $(2 - w)$ gives

$$B'_i + B'_{i+1} = m \cdot w^u \cdot l^v(1 - 2 + l) = m \cdot w^u \cdot l^v(l - 1)$$

thus $B_i + B_{i+1} = B'i + B'i + 1$

case 4: lost-win $\mapsto$ win-lost

let B_i be $-m \cdot w^u \cdot l^v$, then B_{i+1} would be $m \cdot w^u \cdot l^{v+1}$ in case of B_i is a lost; the sum of these two adjacent terms are

$$B_i + B_{i+1} = m \cdot w^u \cdot l^v(l - 1)$$

after the swap, $B'_i = m \cdot w^u \cdot l^v$, and B'_{i+1} would be $-m \cdot w^{u+1} \cdot l^v$. the sum of these two adjacent terms are

$$B'_i + B'_{i+1} = m \cdot w^u \cdot l^v(1-w)$$

we're given

$$w + l = 2 \mapsto w = 2 - l$$

substitude (w) with (2-l) gives

$$B'_i + B'_{i+1} = m \cdot w^u \cdot l^v(1-2+w) = m \cdot w^u \cdot l^v(w-1)$$

thus $B_i + B_{i+1} = B'_i + B'_{i+1}$.

as all cases of adjacent swap could maintain the final result, we could conclude that the final result is independent to the permutation of the cards. □

2. as we're proven that permutation doesn't matter in $\boxed{7.1}$, we can reorder the whole permutation, putting all the wins first, followed by all the lost, written as

$$\begin{aligned}
\text{result} &= \sum_{i=1}^{r} mw^i - \sum_{j=1}^{b} mw^r l^j \text{ where } r = u \text{ and } b = v \\
&= m\sum_{i=1}^{r} w^i - mw^r \sum_{j=1}^{b} l^j \\
&= \boxed{m\left(\frac{w^{r+1}-w}{w-1}\right) - mw^r\left(\frac{l^{b+1}-l}{l-1}\right)}
\end{aligned}$$

3. at $r = 18$ and $b = 20$, let R(w) denotes the final result, with the given r and b:

$$R(w) = m\sum_{i=1}^{18} w^i - mw^{18}\sum_{j=1}^{20} l^j$$

or

$$R(w) = m\left(\frac{w^{19}-w}{w-1}\right) - mw^{18}\left(\frac{l^{21}-l}{l-1}\right)$$

where

$$l = (2-w)$$

so

$$R(w) = m\left(\frac{w^{19}-w}{w-1}\right) - mw^{18}\left(\frac{(2-w)^{21}-(2-w)}{(2-w)-1}\right)$$

to find the maximum gain, with respect to w, we could find the point where $R^{'}(w) = 0$, which gives $\boxed{w = 0.6566}$ and thus $\boxed{l = 1.3434}$

Problem 8: Coupon Bond

A coupon bond with face value F, maturity N years, and annual payment P pays P each year, plus F at maturity. If the annual interest rate is r, how much should you pay for the bond today?

Solution

if we don't accounted for the interest rate, we'll get

$$result = F + NP$$

with interest rate r,

$$result = \frac{F}{(1+r)^N} + \sum_{i=1}^{N} \frac{P}{(1+r)^i}$$

if we let

$$S = \frac{P}{1+r} + \frac{P}{(1+r)^2} + \cdots + \frac{P}{(1+r)^N}$$

and

$$\frac{S}{1+r} = \frac{P}{(1+r)^2} + \frac{P}{(1+r)^3} + \cdots + \frac{P}{(1+r)^{N+1}}$$

their difference gives

$$\frac{S}{1+r} - S = \frac{P}{(1+r)^{N+1}} - \frac{P}{1+r}$$

which we could format it as

$$S = \frac{\frac{P}{(1+r)^{N+1}} - \frac{P}{1+r}}{\frac{1}{1+r} - 1}$$

plug this into our result (with interest)

$$result = \frac{F}{(1+r)^N} + \left(\frac{\frac{P}{(1+r)^{N+1}} - \frac{P}{1+r}}{\frac{1}{1+r} - 1} \right)$$

which simplifies to

$$\boxed{result = \frac{F}{(1+r)^N} + \frac{P}{r}\left(1 - \frac{1}{(1+r)^N}\right)}$$